AIRCRAFT PRODUCTION TECHNOLOGY

V Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIA	SEE	Total
A6AE18	PCC	3	-	-	3	40	60	100

UNIT-I CASTING & WELDING TECHNIQUES IN AIRCRAFT

General principles of various Casting Processes - Sand casting, die-casting, centrifugal casting, investment casting and shell molding types. Principles and equipment used in arc welding, gas welding, MIG welding,TIGwelding, Electrical resistance welding, Laser welding , Electron Beam welding, Soldering and brazing techniques.

UNIT-II MACHINING AND FORMING

General Principles (with schematic diagram only) of working and types-lathe, shaper, milling machines, grinding, drilling m/c, CNC machining. Sheet metal operations-shearing, punching, super plastic forming and diffusion bonding. Bending, Automation in bend forming and different operations in bending like stretch forming, spinning, drawing etc

UNIT-III UNCONVENTIONAL MACHINING

Principles of working and applications of abrasive jet machining, ultrasonic machining, electric discharge machining, electro chemical machining, laser beam, electron beam and plasma arc machining

UNIT-IV HEAT TREATMENT AND SURFACE COATINGS

Heat treatment of Aluminum alloys, titanium alloys, steels, case hardening, Initial stresses and the stress alleviation procedures. Corrosion prevention, protective treatment for aluminum alloys, steels, anodizing of titanium alloys, organic coating, and thermal spray coatings, high temperature coatings-ablative coatings

UNIT-V JIGS & FIXTURES AND INSPECTION TECHNIQUES

Aircraft Tooling Concepts, Types of Jigs, fixtures, wing assembly, fuselage assembly and engine assembly process- tools and equipment for riveted joints, bolted joints, NDT techniques used in aircraft inspection- Dye Penetrant Test, X - ray, magnetic particle and ultrasonic testing.

Text Books:

1. Kalpakjianserope (2011), Manufacturing Engineering and Technology, 5th edition, Pearson Education, New Delhi, India.
2. Manufacturing Technology, P.N. Rao,TMH ,R.K.Jain

Reference Books:

1. Keshu S. C, Ganapathy K. K (2012), Air craft production techniques, E-book, Interline Publishing House, Bangalore.
2. P. C. Sharma (2011), Manufacturing Technology - I & II, 1st edition, S. Chand & Company Ltd. New Delhi

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

At the end of the course the students are able to

1. Compare between primary shaping and joining processes
2. Analyze the working of various material removing techniques by conventional machining processes
3. Compare the working of various material removing techniques by unconventional machining processes
4. Analyze the various heat treatment technique used in aerospace industries
5. Determine the methods used to identify the manufacturing defects.